

YIFAN ZHU

Northwestern Polytechnical University | Software Engineering Undergraduate,
Junior Year

@ laurent.zhu666@mail.nwpu.edu.cn
laurent-zhu.github.io

19200831198

Laurent-Zhu

Xi'an, Shaanxi



EDUCATION

B.Eng. in Software Engineering
Northwestern Polytechnical University

2023.09 – Present

- Academic Performance: GPA 3.813/4.1, ranked 11/270
- English Proficiency: CET-4 613; CET-6 587; CET-6 Speaking Test: Excellent
- Awarded the National Scholarship for Undergraduate Students for two consecutive years

RESEARCH EXPERIENCE

BrepLLM: Enabling Large Language Models to Understand Boundary Representations
ECCV 2026 Conference Submission, Fifth Author

Unified Multimodal LLM Framework for CadQuery-based CAD Generation, Understanding and Editing

Ongoing Research Project

- Built high-quality CadQuery datasets for CAD generation, understanding, and editing tasks; designed unified SFT data formats.
- Conducted model fine-tuning, automatic evaluation, and experimental validation for multimodal CAD modeling.

INTERNSHIP EXPERIENCE

Shenzhen Kuakua Jingling Technology Co., Ltd.

Back-end Developer (Intern)

2025.07 – 2025.08

SELECTED HONORS AND AWARDS

International Level

- Honorable Mention, 2026 Mathematical Contest in Modeling (MCM) (2026.05)

National Level

- National Scholarship for Undergraduate Students (2024.12)
- National Scholarship for Undergraduate Students (2025.12)
- Second Prize, Special Track of the 19th "Challenge Cup" National College Students' Extracurricular Academic Science and Technology Works Competition (2024.11)

Provincial / Regional Level

- Third Prize, 16th National College Students Mathematics Competition, Non-Mathematics Category A (2024.12)
- Third Prize, Northwest Regional Contest, 18th National Undergraduate Software Innovation Competition – Software Design Innovation Contest (2025.04)
- Third Prize, Shaanxi Division, 2025 11th National College Students Statistical Modeling Competition (2025.05)
- Second Prize, Shaanxi Division, 2025 Contemporary Undergraduate Mathematical Contest in Modeling (2025.11)

PROJECT EXPERIENCE

Dual-modal UAV Urban Traffic Perception and Intelligent Analysis System

Undergraduate Innovation Training Project

- Developed a visible-light and infrared UAV traffic perception system using a customized YOLOv8-OBb fusion model on the DroneVehicle dataset.
- Built an Edge-Cloud-Dashboard architecture with WebSocket streaming, FastAPI services, and React-based real-time visualization.